

AYMANE AIT DADS

AI Engineering Intern | LLM-Powered Tools · API Integration · Investment Research Automation

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by engineering an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer - demonstrating production-grade LLM fine-tuning at scale. Reduced manual network analysis time by 60% at Orange Maroc by building automated ML pipelines on 100K+ records, delivering actionable outputs directly to the network optimization team. Combines deep ML engineering fluency - PyTorch, Hugging Face Transformers, LoRA, RAG, LangChain, REST APIs, and agentic pipelines - with hands-on experience integrating AI tools into data-driven workflows. Aims to build LLM-powered research tools at BIT Capital that automate and enhance investment workflows by connecting structured financial data to intelligent, API-driven AI systems.

CORE COMPETENCIES

LLM-powered tools | API integration | workflow automation | investment research | agentic pipelines | database integration | prompt engineering | data infrastructure

WORK EXPERIENCE

Orange Maroc Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built an **automated ML pipeline** combining Random Forest and K-Means on 100K+ fiber-optic network records, eliminating roughly 60% of manual analysis time for the network optimization team.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, producing outputs adopted directly by network engineers for infrastructure decisions.
- Engineered feature extraction and model evaluation workflows in Python using **Scikit-learn and Pandas**, enabling reproducible, scalable analysis across thousands of network nodes.
- Delivered stakeholder presentations translating model outputs into **maintenance recommendations**, bridging the gap between data science findings and business decision-making using Power BI dashboards.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge Kaggle Competition · In Progress (Jun 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) targeting all attention projections, MLP layers, and lm_head.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers to produce 605+ verified chain-of-thought rows.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class leaderboard** (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head and dual learning rates, then submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark.
- Built a **3-model ensemble** with 10-view TTA (flips, crops, blur, resizes) trained on 250K images spanning 25+ generator types, using FP16 mixed precision and AdamW on a T4 GPU.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Statistics, Web Applications, Image Security

CPGE - Mathematics & Physics - Ibn Timiya 2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Hugging Face Transformers | Unsloth | TRL | PEFT | Scikit-learn | vLLM

LLM & AI Tools: LoRA | SFTTrainer | RAG | LangChain | agentic pipelines | prompt engineering | Claude API

Data & Infrastructure: Python | SQL | REST APIs | Pandas | PostgreSQL | Docker | Power BI | Git